MedBridge Patient Agent

Developer Master Document

Google Cloud × ADK Hackathon | Track 1: Build (Net-New Agents)

Healtour Solutions · Kiron Group Ecosystem

Submission Deadline: June 5, 2026, 5:00 PM PT

1. Problem Statement

Medical tourism is one of the fastest-growing healthcare verticals globally, yet the patient experience from inquiry to arrival remains deeply fragmented. A patient seeking treatment abroad must independently navigate hospital selection, cost comparison, document preparation, visa logistics, and travel — typically across 6 to 10 disconnected touchpoints, often with no single trusted coordinator.

The core pain points are:

- No single intelligent agent coordinates the full patient journey end-to-end
- Hospital matching is manual, opaque, and slow — driven by commission, not clinical fit
- Cost estimation is inconsistent; patients routinely encounter hidden charges on arrival
- Document and visa preparation is a recurring failure point, causing missed treatment windows
- Existing platforms provide static information, not dynamic, context-aware guidance

The result: patients either abandon their treatment plans out of friction, or proceed poorly informed — with negative outcomes for both the patient and the healthcare provider.

2. Solution — MedBridge Patient Agent

MedBridge Patient Agent is an autonomous AI agent that coordinates the complete medical tourism journey for an international patient — from first inquiry to pre-arrival readiness — using Google ADK, Model Context Protocol (MCP) tooling, and the Claude Sonnet reasoning engine.

The agent ingests a structured patient intake (condition, country of origin, budget, timeline) and autonomously executes across three MCP-connected tools:

MCP Tool	Input	Output	Owner
Hospital Match	Patient profile + condition	Ranked hospital list from 22-MoU network	Shyam / Ram
Cost Estimator	Hospital + procedure + origin	Itemised cost estimate (INR / USD)	Venkat / Gopi
Document Checklist	Country of origin + hospital	Pre-arrival document list (visa, records)	Pallavi / Hari

The agent synthesises all three tool outputs into a single, structured Patient Journey Plan delivered as an interactive UI response — hospital recommendation with rationale, cost breakdown, and a personalised pre-arrival document checklist.

Healtour Solutions' existing network of 22 hospital MoUs — including Apollo, Fortis, Yashoda, Rainbow, Rela, and Gleneagles BGS — serves as the ground-truth knowledge base for hospital matching. This is real institutional data, not synthetic content. That is the moat.

3. Architecture Overview

The system follows a three-layer architecture: a React frontend, an ADK agent orchestration layer, and three MCP tool servers. All layers are deployed on Google Cloud Run.

Flow summary:

- Patient submits intake via React form (condition, origin country, budget, timeline)
- ADK agent receives the intake and constructs a multi-step execution plan
- Agent invokes Hospital Match MCP tool → returns ranked hospital shortlist
- Agent invokes Cost Estimator MCP tool → returns itemised cost projection
- Agent invokes Document Checklist MCP tool → returns visa and records requirement list
- Agent synthesises all outputs → Claude Sonnet generates the Patient Journey Plan narrative
- Frontend renders structured Journey Plan card to the patient

4. Technology Stack

Layer	Choice	Rationale
Agent Framework	Google ADK	Track 1 — judges reward ADK alignment
LLM Reasoning	Claude Sonnet 4 via API	Multi-step medical context, best tool use
MCP Server	Custom Node.js	Fastest to build; 3 tools needed
Frontend	React + Tailwind	Kiron Group standard stack
Hosting	Google Cloud Run	Judges reward GCP; auto-scales to zero
Hospital Data	JSON knowledge base	Powered by 22 real MoU network

5. Team Structure & Ownership

The team is split into two parallel streams. Both teams work simultaneously and integrate on Day 2 evening. This split prevents any blocking dependency between agent logic and frontend build.

TEAM ALPHA — Agent & Backend Lead: Shyam

Google ADK agent · MCP tool servers · Hospital knowledge base · Cloud Run deployment
Members: Ram, Gopi, Pallavi

Task	Due	Owner
▸ Scaffold ADK agent project structure	Day 1	Shyam
▸ Build hospital JSON knowledge base (22 MoUs)	Day 1	Ram
▸ Hospital Match MCP tool — logic + tests	Day 1	Shyam + Ram
▸ Cost Estimator MCP tool — logic + tests	Day 2	Gopi
▸ Document Checklist MCP tool — logic + tests	Day 2	Pallavi
▸ End-to-end agent flow tested with all 3 tools	Day 2	Shyam
▸ Google Cloud Run deployment (backend)	Day 2	Ram
▸ Architecture diagram (final, submission-ready)	Day 3	Shyam
▸ Code cleanup, comments, README	Day 3	Gopi + Pallavi

TEAM BRAVO — Frontend & Submission Lead: Venkat

React UI · Patient Journey Plan renderer · Demo video · Devpost submission package
Members: Hari, Chandrika, Omkar

Task	Due	Owner
▸ Patient intake form component (React)	Day 1	Hari
▸ Form state management + validation	Day 1	Chandrika
▸ Journey output card + hospital recommendation UI	Day 1	Venkat
▸ Basic routing, layout, Tailwind styling	Day 1	Omkar
▸ Wire all 3 MCP tool outputs to UI components	Day 2	Venkat + Hari
▸ Patient Journey Plan full UI — complete render	Day 2	Chandrika
▸ Google Cloud Run deployment (frontend)	Day 2	Omkar
▸ Cross-team integration test with Alpha backend	Day 2	Venkat
▸ Demo video — patient story format (90 sec)	Day 3	Venkat + Omkar
▸ Devpost submission: code, video, arch diagram, demo	Day 3	Chandrika + Hari

6. 3-Day Execution Plan

Day	Team Alpha — Agent & Backend	Team Bravo — Frontend & Submission

Day 1 Jun 3	▸ Architecture diagram finalized ▸ Hospital JSON knowledge base built ▸ ADK + MCP skeleton scaffolded ▸ React shell & form UI wired	▸ Patient intake form component ▸ Form state management ▸ Journey output card design ▸ Basic routing & layout
Day 2 Jun 4	▸ Hospital match MCP tool live ▸ Cost estimate MCP tool live ▸ Document checklist MCP tool live ▸ End-to-end agent flow tested	▸ All 3 MCP tool outputs rendering ▸ Patient Journey Plan UI complete ▸ Google Cloud Run deployment ▸ Cross-team integration test
Day 3 Jun 5	▸ Architecture diagram (final) ▸ Code cleanup & comments ▸ README & submission package	▸ Demo video (patient story format) ▸ Final UI polish ▸ Submission form completion

7. Integration Checkpoint — Day 2 Evening

Before end of Day 2, both teams must verify the following together:

- Alpha backend API is accessible on Cloud Run and returns valid JSON for all 3 tool calls
- Bravo frontend correctly parses and renders all 3 tool responses in the Journey Plan card
- Full patient story flow runs end-to-end: intake → agent → tools → rendered plan
- Demo URL is live and shareable for the video recording

Integration owner: Shyam (Alpha) coordinates with Venkat (Bravo). Any blocker escalated immediately — do not wait until Day 3.

8. Submission Checklist

#	Deliverable	Owner
1	Source code repository (GitHub, public or with access link)	Shyam
2	Architecture diagram — ADK + MCP layers clearly labelled	Shyam
3	Demo video — 90-second patient story format, MP4	Venkat / Omkar
4	Live demo link (Cloud Run URL with login if private)	Omkar / Ram
5	Devpost project page — all fields complete before 5PM PT Jun 5	Chandrika / Hari

9. Known Risks & Mitigations

Risk	Mitigation	Owner
ADK documentation gaps cause Day 1 blocker	LangChain scaffold prepared as parallel fallback; switch by EOD Day 1 if ADK blocks	**Shyam**
Integration breaks on Day 2 evening	Shyam + Venkat on call together; mock API responses pre-built for frontend fallback	**Both Leads**

Video production overruns Day 3	Script written on Day 2; only 90 seconds needed — no elaborate production	**Venkat**
Hospital data insufficient for match logic	22 MoU hospitals pre-loaded as structured JSON on Day 1; Ram owns this	**Ram**

10. Escalation & Decision Authority

- Technical blocker (ADK / MCP): Shyam decides within 2 hours; escalate to Venkat if cross-team
- Stack change (e.g. ADK → LangChain): Shyam + Venkat joint call; decision logged in shared doc
- Submission / Devpost issue: Chandrika primary, escalate to Venkat immediately
- Deadline risk (any deliverable): Both leads flag to Prasanthi by 6PM IST on Day 2

This document is the single source of truth for the MedBridge hackathon build.
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